

Where Safety and Access Collide

Targeting Virginia's Most Dangerous and Underserved Roads for Smarter Transit Investment

By: Nueral Nexus





EXECUTIVE SUMMARY

8%



Problem Overview

How can we target the deadliest 8% of Virginia’s roads to reduce fatalities and improve transit access in underserved areas?



Value to the
COMMONWEALTH

- Invest smart: Focus on high-risk zones using crash severity + access mismatch data
- Enhance safety: Improve speed enforcement, signage & road design
- Map smart: Use real-time dashboards to flag priority action zones
- Save costs: Prevent repeat crashes & streamline emergency response
- Scale up: Apply model statewide or integrate with transit planning



Ideation



01

HEATMAP

We started with a heatmap of crash hotspots and underserved transit zones — letting the data visually guide us toward the state's most urgent problem areas.

02

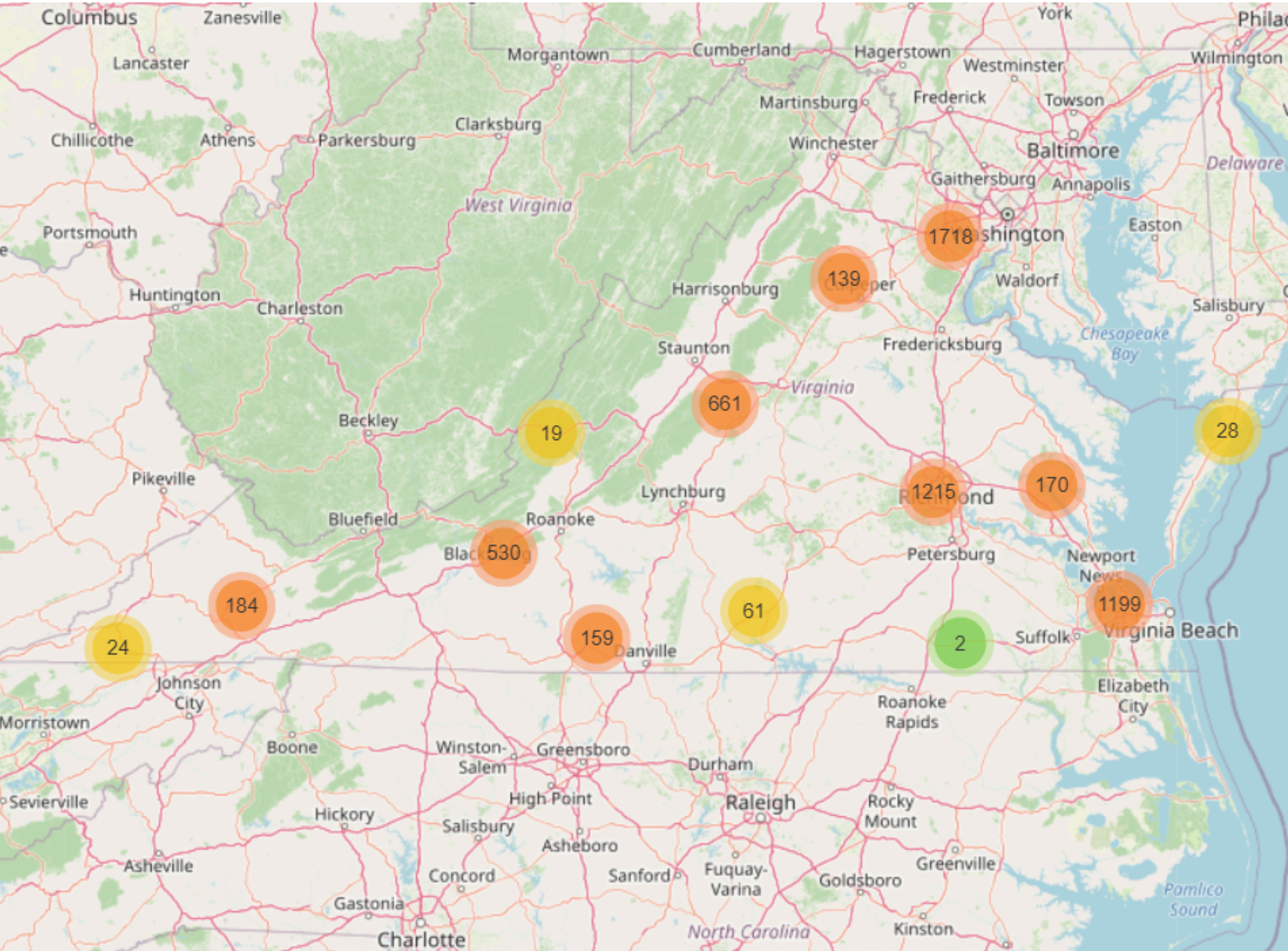
FATALITY TREND SHIFTS

We analyzed how fatal crash involvement has changed over time by vehicle type — revealing which areas are becoming more dangerous, faster.

03

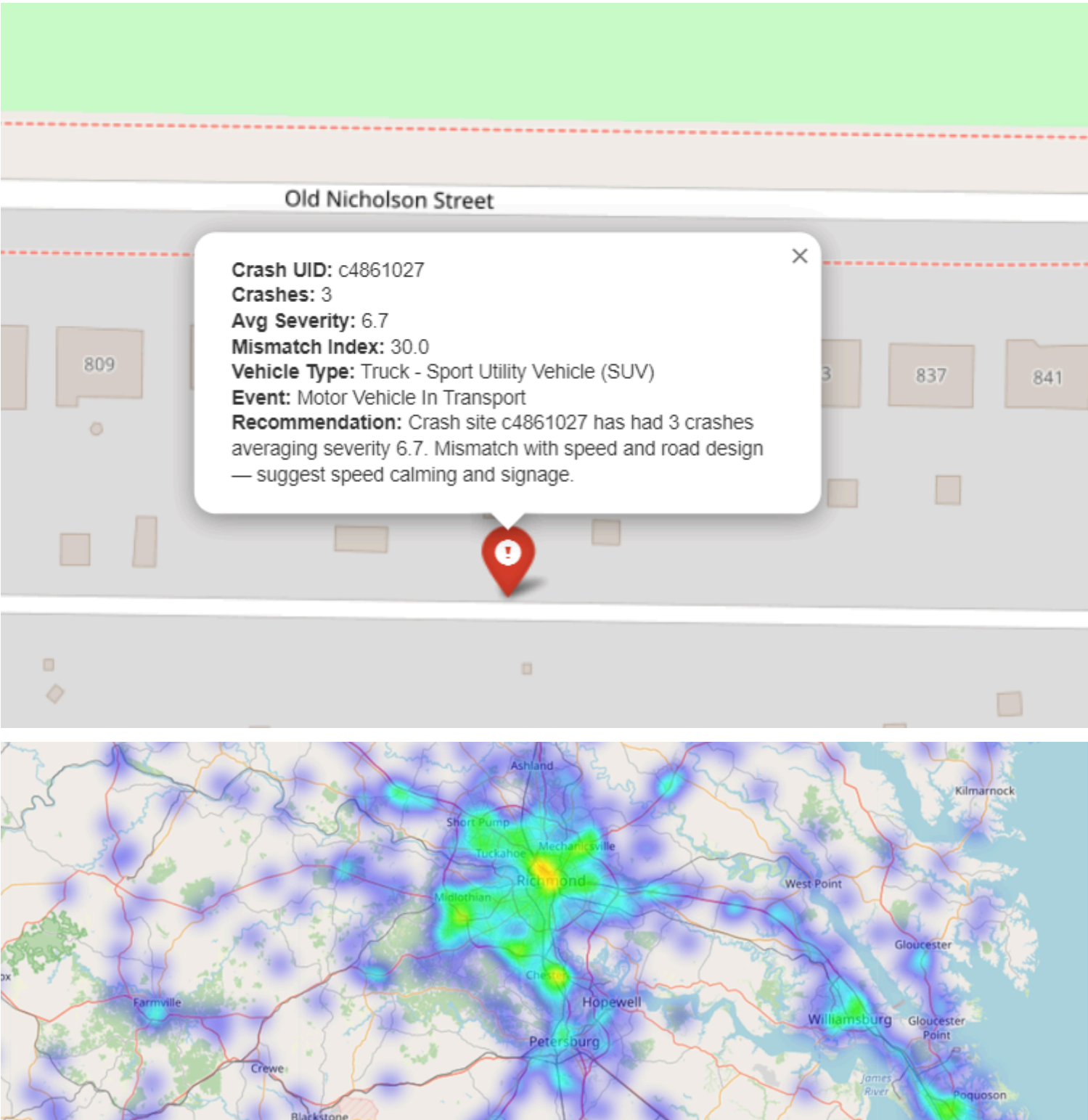
PREDICTION

By combining safety scores, transit coverage, and infrastructure spending data, we built a model to predict where intervention is most needed.



GEOSPATIAL CLUSTERING OF CRASH HOTSPOTS ACROSS VIRGINIA

This map aggregates thousands of crash incidents using clustering to identify regional hotspots.



LOCATION-SPECIFIC SAFETY RECOMMENDATION BASED ON REPEATED HIGH-SEVERITY INCIDENTS

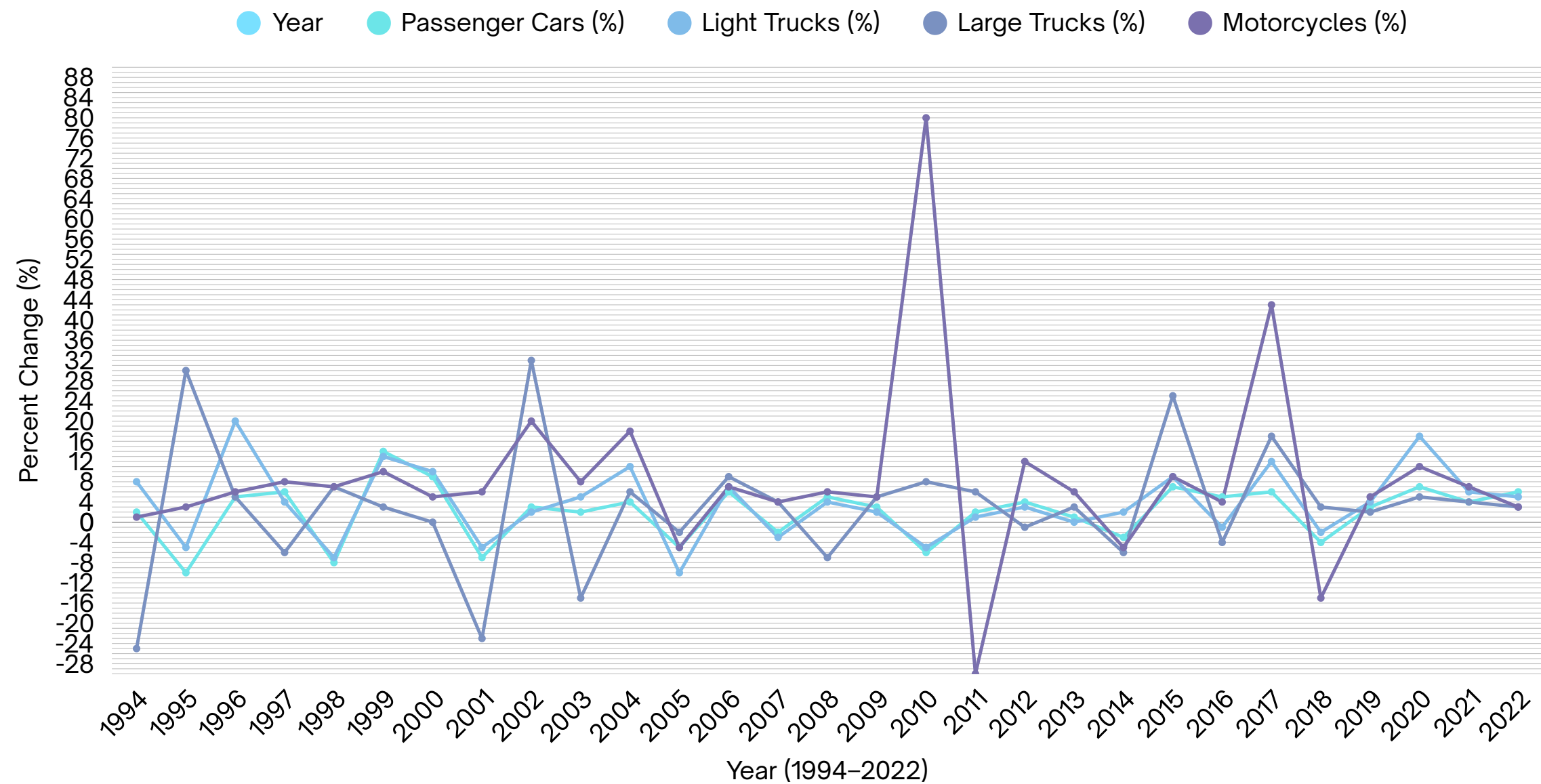
Each marker shows tailored crash data and an AI-generated safety recommendation.

Fatality Trend Shifts

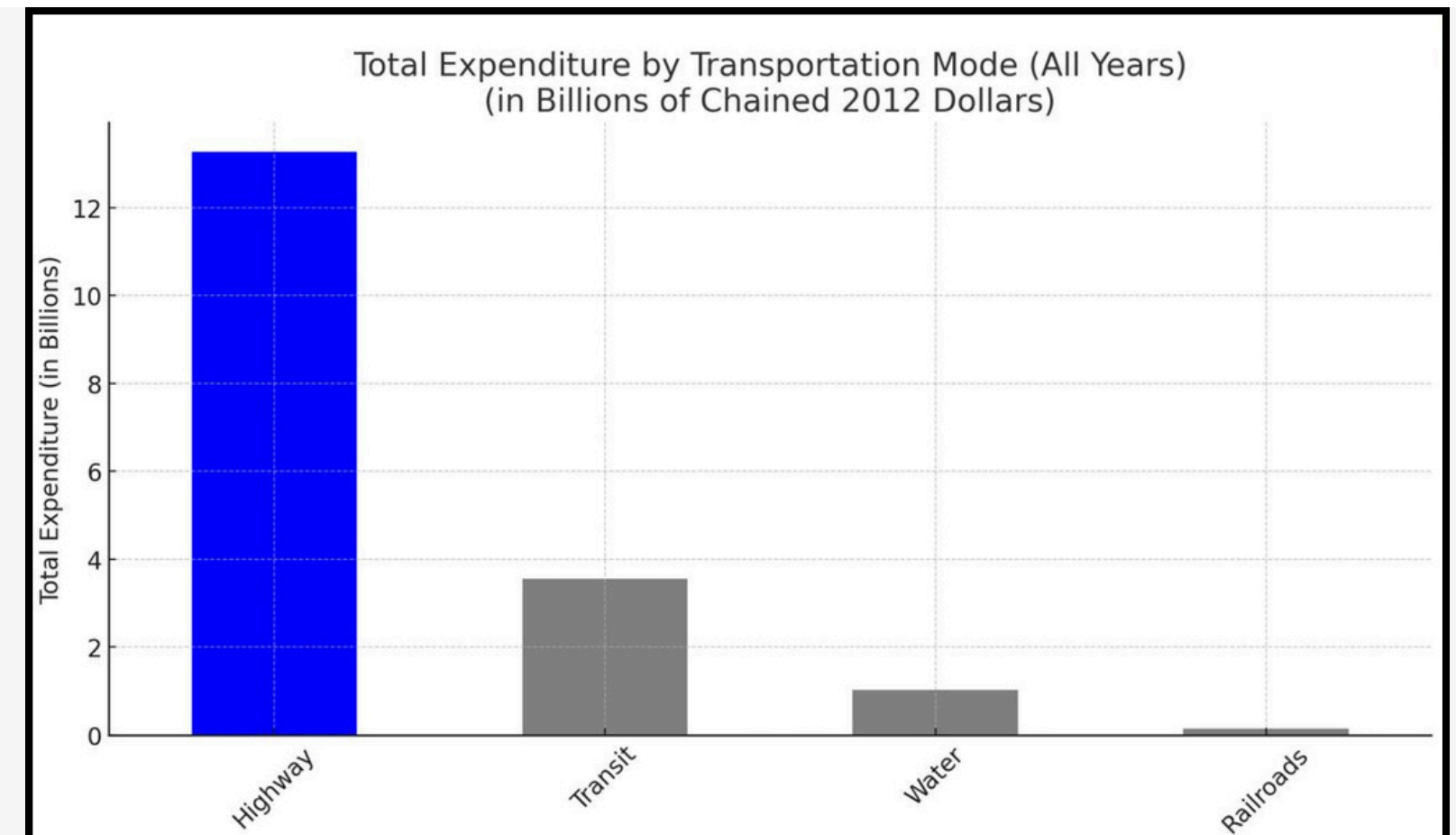
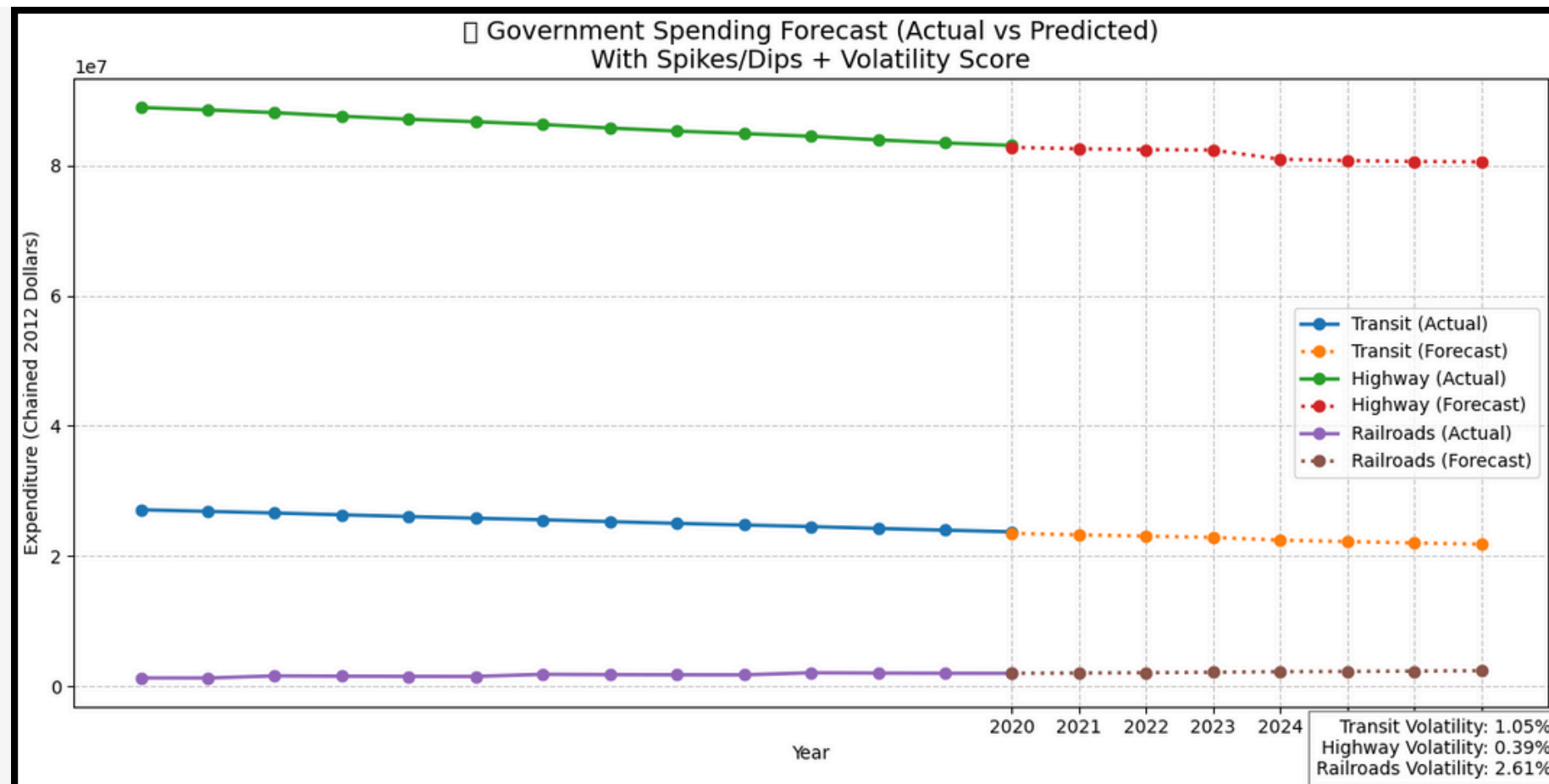
YoY % change in Fatal Vehicle Involvement

FROM 1994 TO 2022

Over nearly three decades, motorcycles consistently exhibit the highest spikes and drops in fatal crash involvement, indicating a volatile and high-risk trend compared to passenger cars, light trucks, and large trucks.



Funding Forecasts Reveal Infrastructure Imbalance



FUTURE SPENDING TRENDS BY MODE

While highway spending shows consistent trends, railroad funding exhibits the highest volatility (2.61%), indicating unpredictable support for alternative transport modes despite long-term infrastructure forecasts.

HIGHWAYS DOMINATE SPENDING

Highway infrastructure accounts for over 70% of total transportation expenditure, significantly overshadowing investment in public transit, rail, and water transport highlighting a major imbalance in multimodal support.

VALUE CREATION



VALUE TO THE COMMUNITY

01 SAFETY MEASURE

Local residents will experience reduced crash frequency and severity. Helps protect vulnerable road users like pedestrians, cyclists, and children near schools and parks.

02 FASTER EMERGENCY RESPONSE

Identifying repeat crash zones allows emergency services to pre-position resources, reducing response time and potentially saving lives.

VALUE TO THE COMMONWEALTH

01 RISK API

The Real-Time Crash Risk API provides dynamic, location-based crash risk data to enhance routing safety for rideshare, delivery, and public transit services.

02 COST EFFICIENCY

Helps the Commonwealth prioritize funding for redesigns, speed control measures, and signage upgrades based on data-driven severity and frequency scoring.

References

[Vehicle Data - Virginia Department of Motor Vehicles](#)

[GTFS Stops - Dec 2024](#)

[GTFS Routes](#)

[Transportation Safety_Fatalities_VA 2010 - 2023](#)

[VDOT 6 Year Improvement Plan - Project list](#)

[Virginia Transit Routes & Stops GTFS - Transit Services](#)

[2019_VTrans_Prioritized_Mid-term_Needs](#)

[Fatal Crashes 1994-2022_VA - National Center for Statistics and Analysis \(NCSA\)](#)

[Traffic Crashes](#)

